Cost and Value Over Time

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We use the tool called discounting to compare money amounts received or paid at different times. Sometimes the problem appears to be one of borrowing and lending, at other times it may look like consumer choice. Still other times it appears as capital budgeting, or options pricing. At the core, all of these problems use the basic machinery of discounting and present values. Every MBA is expected to be skilled at working with discount rates and in solving present value problems.

Present Value

Joanne was promised that she would receive $1000 next year and $1000 more the year after. What is the worth of this promise? A simple answer is $2000. To get a better answer we need to have a way of comparing money today with money next year, and so on. The standard technique is to assume an interest rate and work from there. Suppose we assume that Joanne can earn 7% (per year) on money she invests and holds. How much would she have to have today to have available $1000 next year and another $1000 the year after?

> **In discounting problems we work with the interest ratio, rather than directly with the interest rate. If the interest rate is *r*, the interest ratio is 1+*r*. Joanne's interest ratio is 1.07. That means that if she had $X today it would grow to be $1.07X at the end of one year. Conversely, to have $X at the end of one year, Joanne would have to invest $X/1.07 today.**

Now, let's solve Joanne's problem. The question is the worth of two payments of $1000, one next year and the second the year after. We solve this by asking how much Joanne would have to invest today to replicate these payments (to herself). To pay herself $1000 next year, Joanne would have to invest $1000/1.07 = $934.58 today. The logic is as straightforward as multiplying through by the 1.07 in the denominator: 934.68 x 1.07 = 1000. So, if Joanne put $934.58 to work today at 7%, it would grow to $1000 after one year.

What about the next $1000? Using the same numbers, we can see that if Joanne had a second $934.58 one year from now, she would be able to generate the second $1000 two years from now. To have $934.58 one year from now she needs $934.58/1.07 = $873.44 today. Invest that money at 7% and it will turn into $1000 in two years. Another way of writing this result is like this:

$$873.44 = \frac{934.58}{1.07} = \frac{\left[\frac{1000}{1.07}\right]}{1.07} = \frac{1000}{1.07^2}.$$

So the Present Value of the promise is \$934.58 + \$873.44 = \$1808.02. Having \$1808.02 is equivalent to having \$1000 next year and \$1000 the year after, *if* you have access to a guaranteed 7% rate of return. Exhibit 1 shows how this works in detail.

Exhibit 1
Joanne's Bank Account

Date	Action	Amount	Balance
12/31/1999			0
1/1/2000	Deposit	1,808.02	1,808.02
12/31/2000	Interest Paid	126.56	1,934.58
1/1/2001	Withdrawal	1,000.00	934.58
12/31/ 2001	Interest Paid	65.42	1,000.00
1/1 2002	Withdrawal	1,000.00	0

Generalizing what we just did, the present value of an amount of money X_N received N years from now, discounted at interest rate *r*, is

$$PV = \frac{X_N}{(1+r)^N}.$$

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This is the key formula used in discounting. To discount several amounts of money, we just discount each one separately and sum the results. In Joanne's case, the application of 1 to each of the two payments gives

$$PV = \frac{1000}{1.07} + \frac{1000}{1.07^2} = \$1808.02.$$

An Annuity

Annuities are level payment streams extending over a period of time. Suppose, for example, Joanne were to receive \$1000 every year, "forever." This is called an infinite annuity. What is the PV of that? (For the moment, ignore the fact that Joanne probably won't live forever.) Summing the infinite series isn't hard . . . and the answer is

$$PV = \frac{1000}{r} = \$14{,}285.71.$$

The answer should be obvious in retrospect. The PV is the just exactly the amount of money needed so that 7% interest on it every year is \$1000. If Joanne had \$14,285.71 invested at 7%, she would receive \$1000 per year "forever." The formula generalizes, of course, to any amount of money rather than just \$1000. What is the PV of \$20 per year received forever at a discount rate of 10%? Easy, its 20/0.1 = \$200.

We ignored the fact that Joanne won't live forever. How bad was that assumption? To get a handle on it, we turn to the formula for a finite annuity. What is the present value of \$X received each year, starting one year from now, over N years (N payments)? Skipping over how to work this out, the answer is

$$PV = \frac{X}{r}\left[1 - \frac{1}{(1+r)^N}\right].$$

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Looking at the formula, notice that the term outside the brackets is the value of the infinite annuity. Inside the brackets is an adjustment for the fact that the annuity stops in N years. Usually we don't have to use 2 because there is a simple Excel worksheet function that does this. Use Excel function **PV**(r, N, X) to implement2.

If Joanne is 25 years old and expects to live to be 80, her $1000 per year annuity will last for 55 years. The present value of $1000 received each year for 55 yeas at 7% is given by Excel as **PV**(0.07, 55, -1000) = $13,939.94. So the difference between the value of the 55 year annuity and the infinite annuity is $345.78, about 2.5%. Why is the difference so small? After all, forever is a lot longer than 55 years! Because of discounting. The difference between the finite annuity and the infinite annuity is a delayed infinite annuity−if Joanne were to receive the 55 year annuity and then receive an infinite annuity, it would be the same as an infinite annuity starting now. So the value of an infinite annuity starting 55 years from now is only $345.78. Check it out:

$$PV = \frac{14,285.71}{(1.07)^{55}} = \frac{14285.71}{41.315} = 345.78$$

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That's the power of discounting. $1000 forever has a present value of only $14,286. And that money received 55 years from now has only a value of $346. It seems amazing to most people, but the power of compound interest is all that is at work here. Of course the converse is also true and continues to amaze people. Retirement planning is essentially the insight that if you start early it isn't that hard.

Mortgages and Loans

The finite annuity formula governs mortgages and loans. Given any three of the four variables in 2, the fourth can be calculated. Excel has a whole set of worksheet functions to do just that:

RATE(N,X,PV)[1]	Tells the rate if you know the other variables. Solved by iteration: There is no simple formula.
PV(r,N,X)	Gives the mortgage or loan amount from 2. Reverses the sign of PV relative to X.
PMT(r,N,PV)	Is the formula for the periodic payment given the rate, duration, and mortgage amount. Reverses the sign of X relative to PV.
NPER(r,X,PV)	Tells the duration if you know the rate, payment, and amount.

One trick to using these functions is that loans and mortgages are usually discounted monthly. So you have to be careful to make N the number of months and to make *r* the monthly interest rate. Because annual rates are almost always nominal, the stated annual rate is just 12 times the real monthly rate. So an 8% mortgage has an 8/12 = 0.667% monthly rate.

How much will you pay for a $750,000 30-year mortgage at 7.5%? Use Excel to show that =PMT(0.075/12, 30*12, 750,000) = -$5,244.11 (the negative is supposed to show that it is a payment whereas PV is a withdrawal.). That's the monthly payment. Another way of saying this is "At a discount rate of 0.667% per month, the present value of $5244.11 received each month for 30 years is $750,000."

Present Value as a Criterion of Choice

The standard professional approach to comparing (valuing) streams of money is to compare the present values of the streams. In the previous section we looked at constant streams−annuities. But, in general, money streams are not necessarily constant. However, that

[1] If you memorize the order of variables in this formula you know the rest. The order is r,N,X,PV. That ordering is preserved for the parameters in each of the other formulae (that is, remove the dependent variable and you have the parameters on the right).

poses no difficulty. The present value of a stream of payments is just the sum of the present values of each payment. That is, if the payment received in year K is X_K, then that one payment's present value is $X_K/(1+r)^K$. The present value of the whole stream, $K = 1, 2 \ldots N$ is given by

$$PV = \sum_{K=1}^{N} \frac{X_K}{(1+r)^K}$$

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In Excel, the NPV function takes care of this calculation. The specification =**NPV**(0.1,C1:C10) computes the present value of the cash flows contained in cells C1 through C10, using a 10% discount rate.[2]

As an example of using present values to make a decision, consider a duplicating machine having an expected useful life of five years. The manufacturer offers a maintenance contract costing $375 per year. The buyer estimates that his own maintenance people could handle almost all maintenance problems, but that his experience is that the costs would rise each year. Based on experience, he estimates his maintenance costs as rising linearly from $200 in the first year to $600 in the fifth (and last) year of ownership. Should he take the contract or engage in self-maintenance? Exhibit 2 shows the two streams of expense.

Exhibit 2 Maintenance Expenses

Year	No-Contract	Contract
1	200	375
2	300	375
3	400	375
4	500	375
5	600	375

To compare these two streams, we take their present values. But at what discount (interest) rate? Let's see what happens as several different discount rates. Exhibit 3 displays the present values (PVs) of the two streams at 6%, 10%, 15%, and 20% discount rates.

Exhibit 3 Present Values

Discount Rate	No-Contract	Contract
6%	1,636	1,580
10%	1,444	1,422
15%	1,248	1,257
20%	1,089	1,121

Before going further, be sure you understand Exhibit 3. At a simple mechanical level, the first PV is constructed by using Excel's **NPV** function. If the cells containing the No-Contract flows are B2:B6 (in Exhibit 2), then **NPV**(0.06,B2:B6)=1636. And so on. Now go back to definitions. This means that if we had a bank account that paid 6% per year interest, and if we deposited $1636 into that account today, then we could make the payments shown in the second column of Exhibit 4 for each of the next 5 years. At the end of 5 years the account would be empty. Just like Joanne's problem—were we started, using Exhibit 1.

Do we have a 6% account available? The bank only pays 4% on savings. We could invest in a CD and earn a bit more than 6%. Should we use that rate? What if someone has recently earned 80% in the stock market last year, should they use that discount rate?

The long answer is "take the Finance courses. That is what they are about." The short answer is that a company should discount these money flows at the best rate it can earn elsewhere under similar risk conditions. Again, to fully understand the term "risk conditions," you need to study finance. Turning again to the short answer, if the uncertainty in the future cash flows has nothing to do with the economy as a whole or the industry, then the risk is fully diversifiable and you should use a low risk-free rate (T-Bills). If the uncertainty matches the uncertainties for the firm as a whole, you should use the weighted average cost of capital.

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[2] If the cash flows do not occur at even intervals, check out Excel function XNPV. It takes as input a vector of cash flows and a second vector of dates.

Most of the time, real companies use discount rates that are much higher than those coming from finance theory. Today (year 2000), the cost of capital for most industrial firms is about 8%. Yet you will see corporate managements using 15% and 20% discount rates. Many companies set high discount rates with the idea that setting a high "hurdle rate" will increase profits. This is a mistake−high hurdle rates distort decisions and reject perfectly good projects. Others set high discount rates because they really do want to discount the future heavily−for whatever reason, they don't trust numbers about the distant future. I have seen companies take a 10% after-tax cost of capital and double it because, they say, they are going to work with before tax numbers. That is horribly incorrect reasoning. Tax rates cut your cash flow proportionately; discount rates reduce the impact of cash flow exponentially with futurity.

Returning to Exhibit 5, we see that at 6% and 10%, the Contract is less expensive than No-Contract. At 15% and 20%, the No-Contract option is less expensive. This makes sense because the No-Contract option has rising cash flows. As we increase the discount rate, we lessen the impact of these higher cash flows later on. The crossover discount rate is 13.4% in this case−at that discount rate, both options are equally costly. So if the company's discount rate is less than 13.4%, it should sign the Contract.

Comparing Cash Flows with Different Durations

In the "maintenance contract problem," both options had the same duration: 5 years. When the decision alternatives have different durations, new issues arise. Consider the problem facing PriceAndersonHouse. It recruited administrative assistants from a number of schools, but there were two basic patterns. Recruits from Type I schools were more expensive, commanding starting salaries of $30,000. Recruits from Type II schools less expensive, starting at $26,000.

Exhibit 6
Annual Salary Histories ($000)

Year	1	2	3	4	5	6	7	8	9
Type I	30	33	36	40	44				
Type II	26	28	31	34	38	41	46	50	55

However, recruits from School I tended to stay with the firm for only 5 years, moving on to other opportunities. Recruits from School II stayed an average of 9 years. Both types of recruit received regular raises during their careers. Exhibit 6 displays the average pay patterns for recruits from School I and from School II.

The partners felt that the differences in productivity between these two types of recruits could be ignored−differences between individuals far outweighed the differences between the schools. They were interested, however, in the cost difference between the two career tracks. Their discount rate was 8%.

Taking the present value of both expense streams is easy with Excel. Use =**NPV**(0.08, [cell range]) to find the following:

NPV (5 years) Type I Schools:	$142,579
NPV (10 years) Type II Schools:	$230,746

The Type II recruits are more expensive. But *of course* they are, they stay almost twice as long! The simple present value computation is not really useful in comparing these two alternatives. The problem is that the durations differ. Put differently, the comparison in year 6-9 between $0 for Type I and $40-59K for Type II is not accurate. If a Type I person leaves, they have to be replaced. So the company is paying someone to do that work during years 6-9. What numbers

should we put in those four empty cells? If we start to repeat the Type I salaries, assuming another Type I is hired, the series now goes out to year 10. Now the series for Type II needs to be extended, and so on.

There is a simpler way. Rather than extend the series, let's annualize each present value. That is, turn $142,759 into a 5-year annuity at 8%. Using Excel, **PMT**(0.08,5,-142759)=$35,710. That means that a Type I recruit's salary expense is equivalent to an annual *constant* payment of $35,710. That pattern is very easy to extend beyond the fifth year−it just stays constant. Similarly, the 9-year annuity for Type II recruits is **PMT**(0.08,9,-239746)=$34,388. Now we have two numbers that are comparable: $35.7K for Type I and $34.4K for Type II.

So, the Type II recruits are cheaper. Notice that we used present values (or present costs) in getting this answer, but the present value numbers were then inputs to the annuity-value calculation. The key assumption in this analysis was that the patterns will repeat indefinitely, so that the comparison of annuity values makes sense. If you demand present values rather than annuity values, they are easy to get: just divide the infinite annuity value by the discount rate. That is, for Type I the present value of an infinite stream of hires is 35.7/.08 = $446K, and the present value of an infinite stream of Type II recruits is 34.4/.08=$430K.

To summarize: If you want to compare options of different duration, and are willing to assume that each option is repeated indefinitely, then the easiest method is to (1) get the NPV of the cash flows, and (2) convert the PV to an annuity using the PMT function. To get the NPV of the infinite sequence of repetitions, (3) divide the annuity amount by the discount rate.

There is another way of doing this calculation. You could instead use the present value formula for repeated patterns. If a pattern lasting T years has present value V, and the pattern is repeated indefinitely every T years, then the present value of the infinite stream of repeated patterns is

$$PV = \frac{V}{1 - \frac{1}{(1+r)^T}} .$$

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Exercises

Don't look at the answers until you have worked these out.

1. What is the present value of $100 received 10 years from now? Use a discount rate of 9%.
2. What is the present value of $100 received each year for the next 10 years, starting at the end of this year, discounted at 10%?
3. What is the answer to question 2 if the ten payments start immediately instead of one year from now?
4. A 30-year mortgage loan on $600,000 is offered at a nominal interest rate of 7.25%. What is the monthly payment?
5. Joanne would like to have $1,000,000 put aside for retirement by the time she reaches age 65, 40 years from now. How much should she put aside each year to accumulate this sum, assuming a discount (interest) rate of 7%?
6. Joanne believes that her income will grow at about 2% per year over the next 40 years. She would like to put aside each year a fixed fraction of her (growing) income. How much

should she put aside this year to start accumulating the $1 million she wants at the end of 40 years?

7. A company is buying a machine that will last about 5 years. Management estimates that maintenance will cost $100,000 the first year and increase by 10% each year thereafter. The selling company offers a 3-year maintenance contract for $125,000. At the end of the contract, another 3-year contract can be purchased for the same price. If the company's discount rate is 8%, which is the better deal?

Answers

1. $PV = \frac{100}{1.09^{10}} = \frac{100}{2.3674} = \$42.24$.
2. Excel function PV(0.1,10,-100) = $614.46.
3. Multiply by 1.1 to bring it forward one year. 614.46(1.1)=$675.9.
4. Excel **PMT**(0.0725/12, 360, -600000) = $4,093.06.
5. Two approaches. (A) The present value of $1 million received 40 years from now at 7% is $1000000/1.07^{40}$=$66780. The 40-year level annuity associated with this present value is Excel **PMT**(0.07,40, -66780) = $5009 per year. (B) Simply use the Excel **PMT** function with the future value option rather than the present value option (see Excel Help). **PMT**(0.07, 40, , -1000000) = $5009.
6. Set up an Excel worksheet with an amount saved at the end of year 1 (say $1000), growing by 2% each year thereafter. Pay interest on the beginning balance at 7% per year. Use goal seek to find a final balance of $1 million. Find that the answer is $3916.5.
7. Set up a worksheet with the growing payment stream (100, 110, 121, 133.1, 146.41) and get its NPV at 8% to be $480.43K. Convert this to an annuity using PMT(0.08,5,-480.43) = $120.33K. The other stream is already an annuity at $125K. The self-maintenance option is less expensive.